AI Regulation's Influence on Data Center Siting

SIADS 696: Milestone II Project Report

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Repository: github.com/chalseokorom/ai-law-datacenter-siting **Dashboard:** chalseokorom.github.io/ai-law-datacenter-siting
Report: Google Doc

Introduction

The rapid growth of generative AI has created sudden demand for hyperscale data centers, which can use as much electricity as 10,000 to 25,000 households (Young, 2026). Regulation has not kept pace: there are no binding federal energy standards, and state rules vary and are weakly enforced. Lacking rules built for these facilities, municipalities fall back on decades-old zoning law (Young, 2026).

Existing siting models, such as PNNL's CERF-DC (Mongird & Vernon, 2025), score locations on infrastructure and cost: grid access, land value, electricity prices and taxes. Whether AI legislation is associated with where data centers are actually built has not, to our knowledge, been tested. **We ask whether state AI legislation adds anything to the prediction of where US data centers are sited, beyond the physical and economic infrastructure that already explains it.** The unit of analysis is the county: 3,099 counties in 47 states and DC with complete infrastructure data, 244 of which host at least one facility.

Impact. If policy features carry signal, state legislatures and utility commissions are part of siting decisions, and communities can anticipate pressure from their state's policy record as well as from their substations. If they do not, that is useful too: it tells advocates that infrastructure, not statute, currently drives siting, so leverage sits with utilities, utility commissions and local permitting.

Motivation. State AI legislation has grown quickly. Of the 1,907 bills in our dataset, 1,026 were introduced in the 2025 session alone, against 99 in 2019. Data center construction surged over the same period, and we wanted to know whether the two are connected.

Approach. CERF-DC ranks locations with a fixed cost formula that does not learn from where data centers were built, and PNNL does not publish a model of observed siting. We take the same inputs, train models on real facility locations, and add engineered state AI-policy features. PNNL's projected sites then serve as an external benchmark.

- **Supervised (Part A).** Binary classification of whether a county hosts a data center. Four model families (logistic regression, k-nearest neighbors, balanced random forest, XGBoost) are each trained on three nested feature sets (infrastructure; + Census division; + nine state AI-policy features) and scored with nested, state-grouped cross-validation. A state-shuffled permutation test checks any policy gain. A first pass (notebook 04) with a random county split and two raw bill counts reached the same conclusion; we moved to state-grouped validation because state-constant features can be memorized under a random split.
- **Unsupervised (Part B).** K-means and Gaussian mixture models group counties into archetypes on the same infrastructure features, with and without policy features, and the archetypes are compared with existing and PNNL-projected data centers.

Novel contributions. (1) A county table that joins the CERF-DC rasters, the IM3 atlas, the IM3 projections and NCSL legislation, which no public source provides. (2) State policy features engineered from scraped NCSL bill records. (3) A test design that measures policy against a geography-only control and a state-shuffled null, not just a no-information baseline. (4) An external benchmark of PNNL's projections against observed sites and a learned model.

Main findings

- **Policy features don't reliably help.** The best model, XGBoost, scores PR-AUC 0.368 ± 0.064 with policy against 0.360 ± 0.085 with region alone (no skill: 0.079). The +0.008 gain is far inside its fold-to-fold SD (0.041), and real policy profiles do no better than profiles shuffled between states ($p = 0.16$). Land value and grid access drive the predictions; county population, which none of our sources carries, would add far more (0.37 to 0.52).
- **Three stable county archetypes.** K-means finds tax-light/pricey-land, hot-or-dry/grid-remote and grid-close/cheap-land counties (silhouette 0.26, identical across seven seeds) and outperforms a Gaussian

mixture, which finds the same traits. The tax-light type has the highest data center rate (10.3% vs 7.9% overall), but no archetype set, with or without policy, tracks the label (normalized mutual information ≤ 0.005).

- **PNNL follows existing hubs.** Our top 72 counties overlap PNNL's 72 reference-scenario counties far more than chance (15 vs about 2), but a plain count of existing facilities overlaps them more (32): two-thirds of PNNL's picks already host a facility.

Related Work

The closest work is PNNL's IM3 program, which supplies three of our four datasets but projects future builds as an engineering and cost problem with no policy considerations.

1. **CERF (Vernon et al., 2018).** The Capacity Expansion Regional Feasibility model screens power-plant sites by masking unsuitable locations and ranking the rest by least cost. It was built for power generation, where physical constraints dominate siting. *How we differ:* data centers also respond to markets, incentives and possibly regulation, so we learn from observed sites instead of applying fixed rules.
2. **CERF-DC (Mongird & Vernon, 2025).** CERF's extension to data centers masks unsuitable 100 m cells, then ranks clusters of suitable cells on land, capital, energy, tax and interconnection cost, with an optional market-gravity weight. It is deterministic and scenario driven and never learns from where facilities were built. *How we differ:* we train on observed sites, use CERF-DC's own input rasters as features, and add legislative features it was not built to consider.
3. **IM3 Projected US Data Center Locations (Mongird et al., 2026; Steyn, 2026).** PNNL ran CERF-DC under 4 growth scenarios × 5 market-gravity weights, producing 222 to 1,870 sites per scenario across 29 to 37 states. *How we differ:* we treat the projections as a benchmark, not ground truth, and measure where a learned, policy-aware model agrees or disagrees with them.
4. **Young (2026), “Clouds on the horizon.”** An integrative review of data centers and local US governance: outdated zoning, state incentives and non-disclosure agreements shaping deals, and public opposition being bypassed. *How we differ:* the review is qualitative; we test quantitatively whether a state's legislative record is associated with siting outcomes.

This project is not an extension of a Milestone I project.

Data Sources

Five sources feed one county table. Both parts of the project use the same table, so data and feature engineering are described once (Table 1).

Table 1. Data sources.

Source and location	Format	Key variables	Records used	Time period
CERF-DC package data (Mongird, 2025); MSD-LIVE	12 GeoTIFF rasters, 100 m cells, Albers Equal Area (ESRI:102003), 48,000 × 31,500 cells each	Commercial and industrial electricity rate (USD/kWh), composite siting suitability (0/1), distance to substation, land value (USD/sq ft), real and personal property tax, sales tax, mechanical and water cooling fraction	12 rasters → 3,235 county rows; 3,134 of 3,154 counties matched; 3,099 with complete values	Static snapshot (input year not stated per layer)
IM3 Open Source Data Center Atlas v2026.02.09 (Mongird, 2026); MSD-LIVE	GeoPackage and CSV, WGS84	Facility ID, operator, type (building, campus, point), square feet, latitude/longitude, state and county FIPS	1,479 facility rows (1,239 building, 135 campus, 105 point) → 1,462 rows in 244 modeled counties (see Feature engineering, step 2)	Facilities existing as of February 2026; no build dates
IM3 Projected US Data Center Locations v1.1 (Mongird et al., 2026); MSD-LIVE	20 GeoJSON files (4 growth scenarios × 5 market-gravity weights)	Site polygon, growth scenario, market-gravity weight, state, IT power (MW), campus size, total cost, cooling water demand	16,645 projected sites: 222 (low), 328 (moderate), 909 (high), 1,870 (higher growth) per scenario; 37 states	Projected future buildout
NCSL Artificial Intelligence Legislation Database; ncsl.org	HTML tables scraped with BeautifulSoup (2019–2025 summary pages)	State, bill number, title, summary, status, up to 24 policy categories (e.g. energy, taxes, private sector, government use), session year	2,203 scraped rows → 1,907 unique bills; 2,045 status-history listings	2018–2025 sessions; 1,026 of the bills are from 2025
FIPS reference (data.gov) and US Census Bureau	CSV (FIPS, downloaded Oct 5, 2026; 2024 population estimates); shapefile (2024 county boundaries)	State and county FIPS, names, abbreviations; Census division; county population; county polygons	40,411 rows → 3,154 unique counties	2024

Why these sources? The CERF-DC rasters are the inputs PNNL used, so any difference from its projections comes from the learning approach and the policy features, not from different infrastructure data. The atlas is the only open, national, facility-level inventory. It is crowd-sourced from OpenStreetMap, so coverage is uneven.

Outcome label. has_data_center = 1 if the county holds at least one atlas facility. Among the 3,099 counties with complete infrastructure data, 244 are positive (7.9%), holding 1,462 facility rows. Virginia alone accounts for 319 of the 1,479 atlas rows, followed by Texas (127), California (112), Oregon (109) and Ohio (93), so the label is both rare and geographically concentrated.

Feature Engineering

Every source arrives at a different geographic level (100 m cells, facility points, project polygons, state bills). The core of the feature engineering is bringing each one to a single row per county, keyed on a 5-digit FIPS string with leading zeros kept (Figure 1).

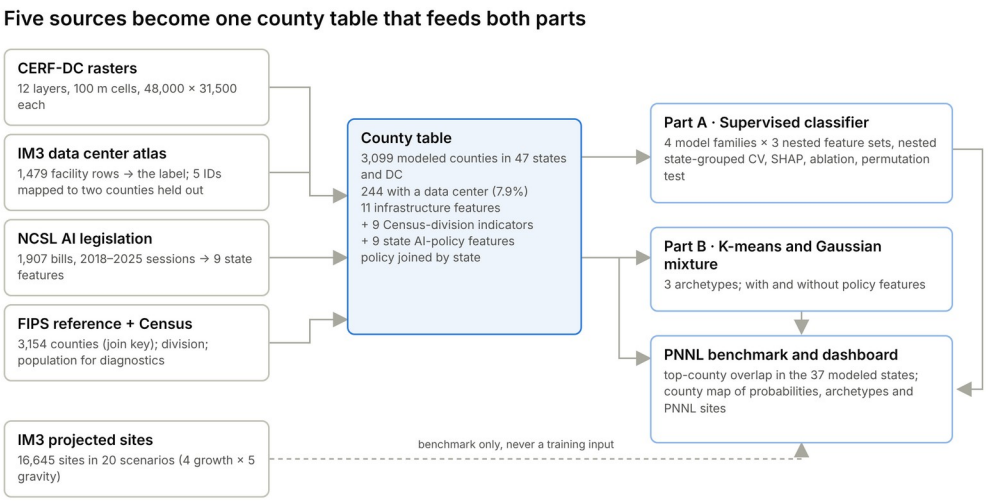


Figure 1. Data pipeline: five sources, one county table, three analyses. PNNL's projections never enter training.

1. **County features from CERF rasters (notebook 01).** We profiled each 1.5-billion-cell raster block by block to stay within memory and confirmed that all 12 share one grid. Zonal statistics over 2024 Census county polygons give the county mean for continuous rates and values, plus the minimum distance to a substation (is any land close to the grid?) and the maximum suitability (is any land suitable?).
2. **Cleaning and joining (notebook 02).** The FIPS reference collapsed from 40,411 rows to 3,154 counties with no conflicting names. Atlas facilities were counted per county. Five facility IDs that the atlas maps to two counties each (10 rows; the atlas notes that these straddle a county line) and seven rows with county codes outside the reference (Connecticut planning regions, Puerto Rico) were written to review files and left out of the counts. Every merge validated its key relationship and reported its match coverage.
3. **Policy features (NCSL pipeline and notebook 05).** A stable bill ID was recovered from each bill's source URL, bills were deduplicated, and statuses were mapped to Passed (325), Failed (707), Pending (857) or Vetoed (18). Topics were standardized to NCSL's 24 categories and grouped into three content areas. The pipeline also built bill-level text, topic (TF-IDF with NMF to find themes, then keyword topics) and status-history features. We aggregate only the nine state features in Table 2: the outcome is a county, so a county inherits its state's record, and every extra state-level column is one more way for a model to recognize states. A state with two bills can only have a 0%, 50% or 100% pass rate, so pass rates and shares are shrunk toward the national value with a prior of five bills: (count + 5 × national rate) / (total + 5).
4. **Geography control.** Each county's Census division is encoded as nine one-hot columns. This is the policy-free baseline that the policy features must beat.

Table 2. State AI-policy features (2018–2025 sessions) and their range across the 48 modeled jurisdictions.

Dimension	Feature	Definition	Range across states
Attention and follow-through	Bills introduced	log(1 + bills introduced)	2 (SD) to 230 (NY) bills
	Bills enacted	log(1 + bills enacted or adopted)	0 (DC) to 26 (TX)
	Pass rate	Passed ÷ decided bills, shrunk toward the national 31%	0.12 (MO) to 0.71 (ND)
Timing	First session year	Year of the state's first AI bill	2018 (NJ) to 2024 (SD)
Content	Regulatory share	Share of bills regulating the private sector, shrunk	0.21 (KS) to 0.72 (NY)
	Public-use share	Share of bills on government or judicial use, appropriations, audit, shrunk	0.12 (MT) to 0.64 (IA)
	Rights share	Share of bills on criminal use, elections, personhood, provenance, shrunk	0.06 (NJ) to 0.45 (KS)
Infrastructure	Infrastructure bills	log(1 + bills tagged energy or taxes, or mentioning “data center”)	0 (30 states) to 15 (NJ)
	Infrastructure bill enacted	1 if any infrastructure bill passed	1 in 7 states

5. **Noisy and missing data.** Two electricity rasters hold impossible values: commercial rates of 73.80–215.47 USD/kWh in 13 South Carolina counties (9 distinct values across 2.94 million cells, a coding artifact) and industrial rates up to 300.04 USD/kWh in Alabama and Colorado. Both rates are capped at the 99th percentile, learned inside each training fold for the supervised models. The market-location raster is zero for every county at the county level and encodes existing facilities, so it is excluded to avoid label leakage. 55 counties without CERF data (Alaska, Hawaii, Connecticut's legacy counties and a few others) and one malformed FIPS row are dropped, leaving 3,099. Water and mechanical cooling fractions sum to 1, so the supervised models keep only mechanical cooling. Appendix B gives the evidence for each issue.
6. **Final features.** The supervised models use 29: infrastructure (11: commercial and industrial electricity rate; mean and maximum suitability; mean and minimum substation distance; land value; real property, personal property and sales tax; mechanical cooling), Census division (9) and policy (9, Table 2). Land value and both substation distances are log-transformed. Clustering uses 11 CERF features: the same infrastructure set without maximum suitability and with the water cooling fraction, standardized. Appendix A lists every feature with its source and statistic.

Part A. Supervised Learning

Methods

Task and feature representation. Binary classification at the county level: does the county host at least one atlas facility (244 of 3,099, 7.9%)? The experiment compares feature sets, not just models, so each family is trained on three nested sets (Table 3). The middle set is the control that matters. Policy features are constant within a state, so any state-level variable can look predictive simply by flagging Virginia or Texas. Policy has to beat a policy-free geographic control to count. We use the nine Census divisions rather than state dummies because, under state-grouped validation, a held-out state has no dummy of its own.

Table 3. Nested feature sets.

Set	Features	What it tests
Infrastructure	11 CERF-DC county features	How far PNNL's own inputs go when learned from observed sites
+ Geography	+ Census division (9 indicators)	How much is explained by where a county is, with no policy content
+ AI policy	+ 9 state AI-policy features (Table 2)	Whether policy adds signal beyond geography

Model families and why. Logistic regression (linear, probabilistic) gives signed coefficients a policy reader can check, with class weights for the 7.9% positive rate. **k-nearest neighbors** (instance based) asks a substantive question: do data center counties sit close together in infrastructure space? **Balanced random forest** (bagged trees on class-balanced bootstraps) captures thresholds and interactions and handles imbalance without synthetic data. **XGBoost** (boosted trees) is usually strongest on tabular data, uses scale_pos_weight for imbalance, and supports exact TreeSHAP. The two tree ensembles are deliberately paired: bagging and boosting fail differently, so agreement between them is evidence about the features rather than about one learner. Logistic regression and k-NN see standardized features; all models see electricity rates capped at the training fold's 99th percentile and log-transformed land value and substation distances.

Validation and tuning. Nested, state-grouped cross-validation: StratifiedGroupKFold with 5 outer folds × 3 seeds (15 test folds), and a 3-fold state-grouped grid search on PR-AUC inside each training split. Counties from one state are never in both train and test. Grids: logistic $C \in \{0.01, 0.1, 1, 10\}$; k-NN $k \in \{10, 25, 50, 100\}$ × uniform or distance weights; random forest depth $\in \{6, 12, \text{none}\}$ × min leaf $\in \{1, 5\}$; XGBoost depth $\in \{2, 3, 5\}$ × learning rate $\in \{0.03, 0.1\}$ × trees $\in \{200, 500\}$. The most frequent winners (+ AI policy set) were $C = 0.01$ (7 of 15 folds), $k = 25$ with distance weights (9), depth 12 with min leaf 1 (4, tied with unlimited depth and min leaf 5), and XGBoost depth 2, rate 0.03, 200 trees (9). These modal settings are then held fixed for the ablation, permutation and sensitivity analyses, so those numbers differ slightly from Tables A1–A2. Because class weighting inflates raw probabilities and the dashboard shows probabilities, scores are calibrated with cross-fitted isotonic regression (each fold mapped by a calibrator fit on the other folds).

Evaluation

Metrics and why. PR-AUC (average precision) is primary: with 92% negatives, ROC-AUC rewards ranking the many easy rural negatives and can look strong for a weak model, while PR-AUC measures how well the positives are ranked (no skill = 0.079). **Brier score** measures probability accuracy, which matters because the dashboard shows probabilities. **ROC-AUC** is reported for comparability. Two reference rankings involve no learning: the positive rate, and CERF's own composite suitability score.

Table A1. Best model per family on the + AI policy set, mean ± SD over 15 state-grouped test folds. Brier uses calibrated probabilities.

Model	PR-AUC	ROC-AUC	Brier
No skill (positive rate)	0.079 ± 0.005	0.500	0.073 ± 0.004
CERF composite suitability ranking	0.189 ± 0.058	0.678 ± 0.047	—
Logistic regression	0.351 ± 0.079	0.788 ± 0.063	0.066 ± 0.004
k-nearest neighbors	0.266 ± 0.080	0.741 ± 0.068	0.071 ± 0.009
Balanced random forest	0.364 ± 0.091	0.799 ± 0.062	0.064 ± 0.007
XGBoost	0.368 ± 0.064	0.801 ± 0.056	0.062 ± 0.004

Table A2. PR-AUC by feature set (mean ± SD, same 15 folds) and the paired policy effect, (+ AI policy) − (+ Geography) per fold.

Model	Infrastructure	+ Geography	+ AI policy	Policy effect	Folds improved
Logistic regression	0.349 ± 0.081	0.360 ± 0.094	0.351 ± 0.079	−0.009 ± 0.030	4 of 15
k-nearest neighbors	0.285 ± 0.104	0.267 ± 0.070	0.266 ± 0.080	−0.000 ± 0.044	6 of 15
Balanced random forest	0.348 ± 0.098	0.353 ± 0.103	0.364 ± 0.091	+0.010 ± 0.039	8 of 15
XGBoost	0.356 ± 0.073	0.360 ± 0.085	0.368 ± 0.064	+0.008 ± 0.041	9 of 15

Reading the tables. Every learner beats CERF's own suitability ranking by a wide margin, so observed sites teach something the engineering rules miss. The tree ensembles lead, consistent with threshold effects such as land value mattering only above a level; k-NN trails, so data center counties are not tightly grouped in infrastructure space. The gap between families (up to 0.10) is far larger than the gap between feature sets within a family (at most 0.019): for this task, model choice matters more than any feature set we added. Fold by fold, adding policy to region moves PR-AUC by −0.009 to +0.010 across families; every effect sits well inside its own SD, and the sign is not consistent.

Feature importance and ablation (XGBoost)

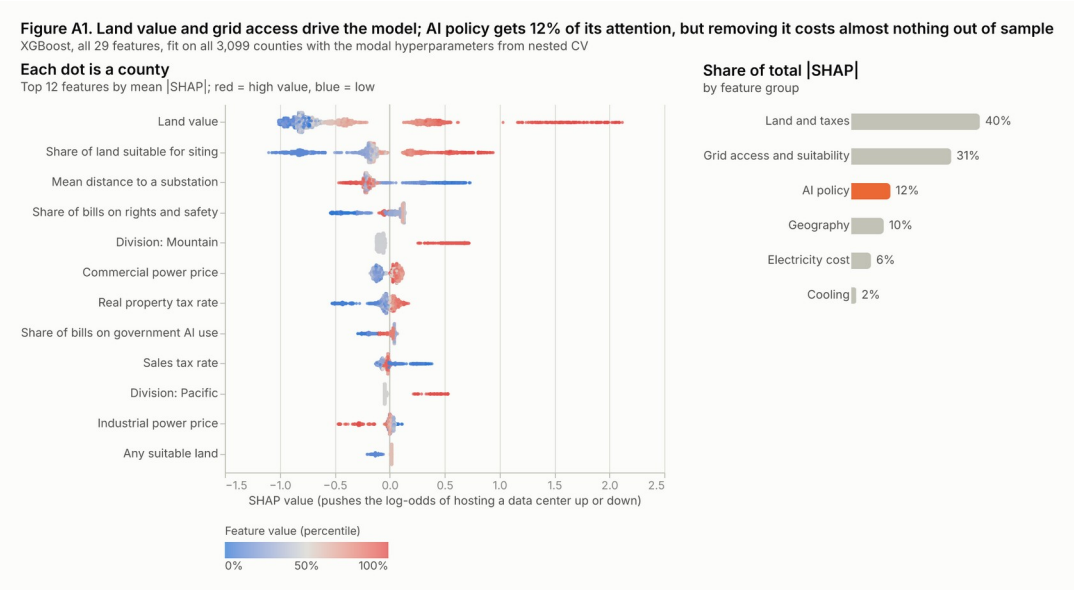


Figure A1. SHAP for XGBoost fit on all counties with the modal hyperparameters. Higher land value, more suitable land and shorter substation distances raise the predicted odds.

Land value dominates, followed by the share of suitable land and the mean distance to a substation. AI-policy features take 12% of the model's total |SHAP|, slightly more than region (10%), led by the share of rights-and-safety bills, yet that feature's direction is weak (Spearman 0.25 between value and SHAP) and removing all nine policy features does not lower out-of-sample PR-AUC (Table A3). The model uses them to tell training states apart, not to generalize. Commercial power price has a positive direction, so it stands in for dense coastal markets rather than measuring cost.

Table A3. Group ablation: XGBoost with fixed modal hyperparameters, PR-AUC mean ± SD over the same 15 folds (all 29 features: 0.371 ± 0.067). *Removed* drops one group; *alone* trains on that group only.

Feature group	Features	Removed	Alone
Electricity cost	commercial and industrial rate	0.378 ± 0.067	0.086 ± 0.022
Grid access and suitability	substation distance mean/min, suitability mean/max	0.303 ± 0.052	0.258 ± 0.094
Land and taxes	land value; real, personal property and sales tax	0.264 ± 0.088	0.326 ± 0.069
Cooling	mechanical cooling fraction	0.376 ± 0.070	0.093 ± 0.014
Geography	Census division (9 indicators)	0.359 ± 0.061	0.125 ± 0.024
AI policy	9 state features (Table 2)	0.372 ± 0.074	0.100 ± 0.027

State-level permutation test. We shuffled the nine policy columns across states 100 times (each state keeps one internally consistent profile, but it now belongs to another state) and refit with the same folds and hyperparameters. Shuffled profiles cost XGBoost 0.007 PR-AUC on average, the price of extra state-level noise; the real profiles cost 0.001, better than 85 of the 100 shuffles ($p = 0.16$). For logistic regression the real profiles do no better than chance (−0.008 vs −0.012; $p = 0.36$). Added one dimension at a time, no policy dimension changes XGBoost by more than ±0.005 (timing +0.004, infrastructure bills −0.005). **Insight into features:** policy carries no information about siting beyond what an arbitrary state-level variable would; the signal lives in land value and grid access.

Sensitivity analysis

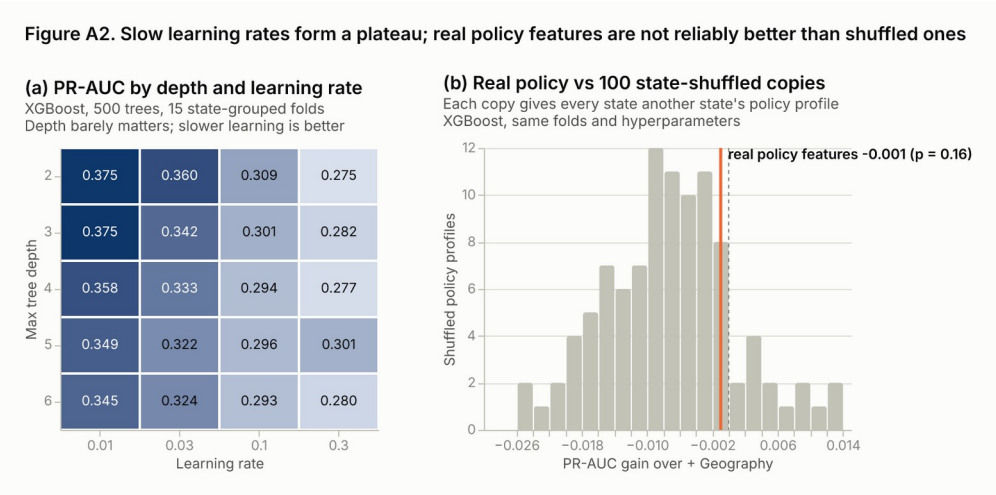


Figure A2. (a) Hyperparameter sensitivity: depth barely matters, the learning rate does (PR-AUC 0.275 to 0.375). The nested search chose the same region (depth 2–3, rate 0.03, 200 trees). (b) The real policy features sit inside the null distribution of 100 state-shuffled copies.

Table A4. Solution-element sensitivity: XGBoost, fixed hyperparameters, PR-AUC on the + AI policy set, with the policy effect over + Geography in brackets.

Variation	PR-AUC (policy effect)
Electricity capped at the 99th percentile (default)	0.371 ± 0.067 (−0.001)
Electricity outliers left raw	0.374 ± 0.068 (+0.001)
Commercial electricity dropped	0.371 ± 0.064 (−0.004)
Random (not state-grouped) cross-validation	0.377 ± 0.085 (−0.004)
+ county population and density (demand proxy, outside our design)	0.522 ± 0.052 (−0.008)

The policy effect stays between −0.008 and +0.001 under every change. Trees are indifferent to the raw electricity artifacts. A random split barely flatters this shallow XGBoost (0.377 vs 0.371), but it lifts logistic regression with policy from 0.346 to 0.372, the state memorization our design guards against. The last row is outside the design: two Census variables raise PR-AUC by 0.15, far more than any policy feature.

Tradeoffs

- **Precision vs recall.** For community preparedness, recall matters more: missing a county that gets a data center is costlier than reviewing a false alarm. For investment screening, precision matters more. Table A5 shows both sides.
- **Interpretability vs accuracy.** Logistic regression trails XGBoost by only 0.017 PR-AUC with policy and 0.007 on infrastructure alone, so a readable linear model loses little; the gap opens only when policy is added (with region alone both score 0.360).
- **Honest validation vs headline score.** State-grouped CV costs logistic regression with policy 0.026 PR-AUC relative to a random split, and notebook 04's single random test split scored higher still (0.46), but only the grouped score says anything about an unseen state.
- **Calibration vs ranking.** Class weighting improves the ranking of rare positives but inflates probabilities; isotonic calibration repairs them at the cost of a held-out step.

Table A5. Precision, recall and counties flagged at three thresholds (XGBoost, + AI policy, calibrated out-of-fold probabilities, mean over three seeds).

Threshold	Precision	Recall	Counties flagged
0.05	0.179	0.758	1,040
0.20	0.362	0.426	288
0.50	0.495	0.124	62

Failure analysis

We chose one county per failure category by rule from the out-of-fold predictions (XGBoost, + AI policy): the highest-scoring county with no atlas facility, the lowest-scoring county with an atlas campus, and the largest shift toward the wrong answer when policy is added. Each is explained with the three fold models that held out its state (Figure A3).

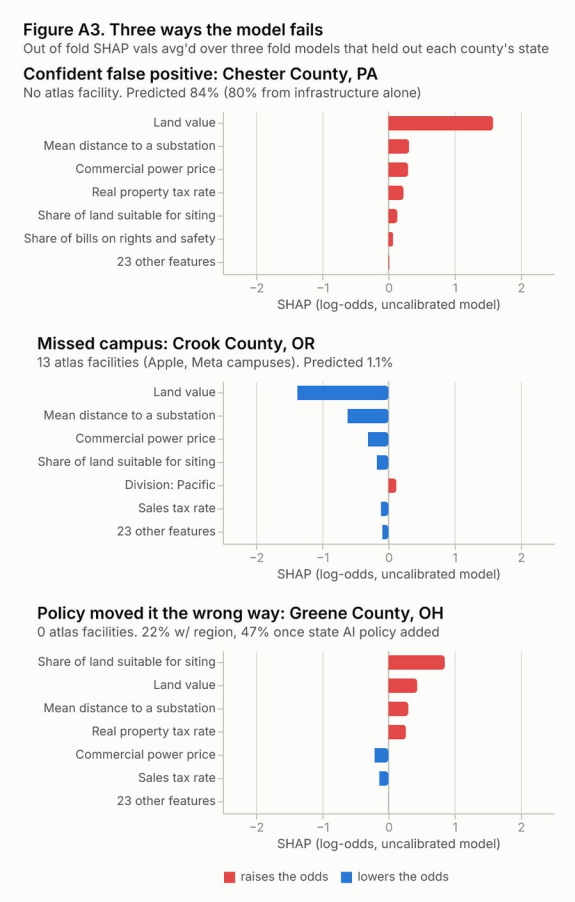


Figure A3. Out-of-fold SHAP contributions (log-odds of the uncalibrated model) for the three cases.

1. Confident false positive: Chester County, PA (42029). Predicted 84% (80% from infrastructure alone); no atlas facility. Its land value is in the top 5% of counties and its mean substation distance in the shortest 16%, so the model reads this Philadelphia suburb (560,745 people) as a data center county; neighboring Delaware and Lancaster counties are the next false positives. *Category:* systematic proxy error with possible label noise. Land value stands in for urban demand, and an OpenStreetMap-based atlas can miss small facilities, so we treat it as a review candidate. *Fix:* add demand and market features (population, fiber, distance to existing hubs) and check high-scoring negatives against a second inventory.

2. Missed campus: Crook County, OR (41013). Predicted 1.1%; the atlas holds 13 records here, including Meta's and Apple's Prineville campuses (largest 17.4 million sq ft). Every top feature pushes down: land value in the bottom 11%, mean substation distance in the longest 3%. *Category:* edge case, rural hyperscale outlier. Campuses this size chase cheap power, tax deals and land at a scale that county averages do not capture. *Fix:*

facility-scale features (utility tariffs, enterprise zones, transmission capacity) or a separate model for campus-scale siting.

3. Policy moved it the wrong way: Greene County, OH (39057). No atlas facility. Region gives it 23%; adding state AI policy lifts it to 47%. The policy features' own SHAP contribution is small (+0.12 log-odds, led by Ohio's rights-and-safety share); most of the shift comes from how the policy model re-weights Greene's strong infrastructure (suitability in the top 5%, land value in the top 13%). Across Ohio the median shift is only +0.006. *Category:* instability from state-level features. Nine columns that are constant within a state change the trees for every county in it, and a few counties near a decision boundary swing widely. *Fix:* county-level policy (moratoria, zoning, local abatements) instead of one statewide profile, and regularization that limits splits on state-constant features.

Beyond the three cases, we tested six failure hypotheses fixed before the models ran (Appendix Table A6). Missing demand is the main failure: the largest population quintile is under-predicted (observed 30.5%, predicted 20.6%), the three smallest are over-predicted (1.0% vs 4.0%), and 35 of the 52 counties with a facility of at least 1 million sq ft are missed at a 0.20 threshold. Policy moves whole states together, but only 4.3% of residual variance lies between states, and the raster artifacts are not a material error source.

External benchmark: PNNL projections

Within the 37 states PNNL modeled, we compare its 72 reference-scenario counties (moderate growth, 50% market gravity; 328 sites) with our 72 highest-probability counties. All three of our feature sets overlap PNNL far more than the 2 counties expected by chance (13, 13 and 15 with policy), while CERF's own suitability raster overlaps least (9). A plain count of existing facilities overlaps most (32), and 67% of PNNL's chosen counties already host a facility. PNNL-only picks are smaller established hubs our model scores low (Mecklenburg, VA, 0.02; Grant, WA, 0.06; even Loudoun, VA, scores only 0.22); our-only picks are dense metro counties with pricier land and no recorded facility. PNNL's projections largely follow existing hubs, while our model follows urban land markets.

Part B. Unsupervised Learning

Methods

Goal and feature representation. Find county archetypes from infrastructure alone, without the label, then check which archetypes hold existing and projected data centers. Clustering answers a question the classifier cannot: are there distinct kinds of viable county rather than one ranking? We use the 11 CERF features (Feature engineering, step 6), with both electricity rates capped at the 99th percentile and all features standardized, because K-means uses Euclidean distance and raw units differ by four orders of magnitude (tax rates near 0.01, substation distance near 77).

Method 1: K-means (non-probabilistic, centroid based). Assigns each county to the nearest of K centroids. Chosen because a choropleth needs one clear archetype per county, and centroids read directly as profiles.

Method 2: Gaussian mixture (probabilistic). Models the data as K Gaussians, each with its own covariance, and returns a membership probability per county. Chosen because it differs from K-means in exactly the way that matters here: K-means forces spherical clusters and hard boundaries, while a mixture allows elongated clusters and gives a confidence per county.

Hyperparameter exploration. K-means: K from 2 to 10 with 10 initializations, on raw and capped data, judged by inertia (elbow), silhouette and smallest cluster (notebook 03, Figure B1); stability across seven seeds and after capping three more long-tailed features. Mixture: K from 2 to 10 × full or diagonal covariance, 5 initializations, judged by BIC, silhouette of the hard labels and smallest component, and reg_covar from 0.001 to 1. Notebook 03 first fit a four-component mixture (silhouette 0.219, smallest component 239); notebook 09 refits it at K = 3 with a larger covariance floor for a like-for-like comparison.



Figure B1. Choosing K. On raw data the top silhouette (0.399 at K = 9) comes from clusters that isolate single counties with impossible electricity rates; on capped data K = 3 is the largest K without a near-empty cluster.

Evaluation

Metrics and why. Silhouette (cohesion vs separation, −1 to 1) and Davies-Bouldin (lower is better) work for both methods without a label. Inertia and BIC trace model fit as K grows; BIC penalizes extra components. Smallest cluster size catches solutions that look good only because they isolate outliers. The adjusted Rand index (ARI) measures agreement between two clusterings and is our stability metric. Normalized mutual information (NMI) with the label, the state and the Census division tests what the archetypes actually encode.

Table B1. Best model per method on the same 11 features. Silhouette and Davies-Bouldin on all 3,099 counties.

Method	K	Silhouette	Davies-Bouldin	BIC	Smallest cluster	ARI vs K-means
K-means (capped CERF)	3	0.263	1.50	—	651	1.000
Gaussian mixture, full covariance	3	0.216	1.72	14,813	681	0.717
Gaussian mixture, BIC minimum (full)	10	0.126	2.23	−7,684	29	0.373

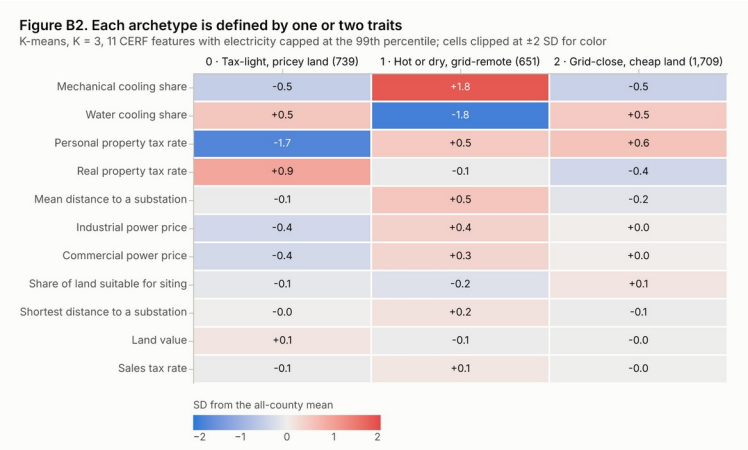


Figure B2. K-means archetype profiles in standard deviations from the all-county mean.

Two archetypes are each defined by one trait. **Tax-light, pricey land** (739 counties) has no personal property tax (−99% vs the mean), higher real property tax (+46%) and land value (+82%), and cheaper power (−13 to −17%). **Hot or dry, grid-remote** (651) must cool mechanically (+334%; the arid West and the humid Southeast alike), sits far from substations (minimum distance +135%) and pays more for power. **Grid-close, cheap land** (1,709) is the remainder, with the most suitable land (+23%). Cluster 0 is defined almost entirely by personal property tax, which CERF-DC sets at 0.0125 or 0 by state: K-means partitions counties along a state tax-policy line before any AI-policy feature is added.

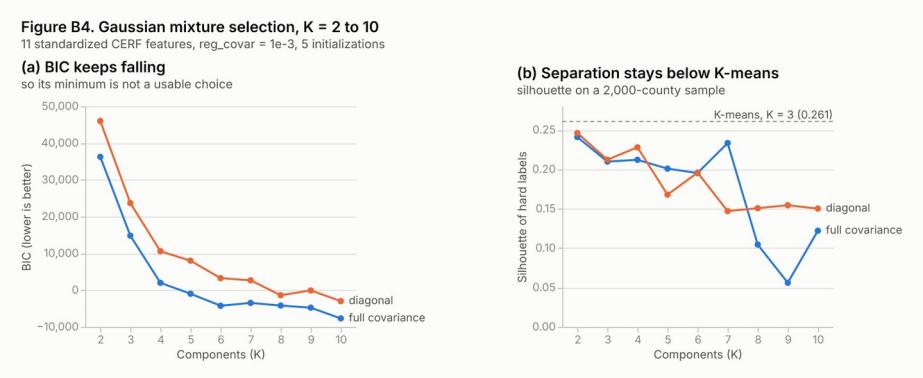


Figure B4. Gaussian mixture selection. BIC falls almost steadily to K = 10, because binary tax rates and cooling fractions piled at 0 and 1 reward extra components; no mixture beats K-means on silhouette.

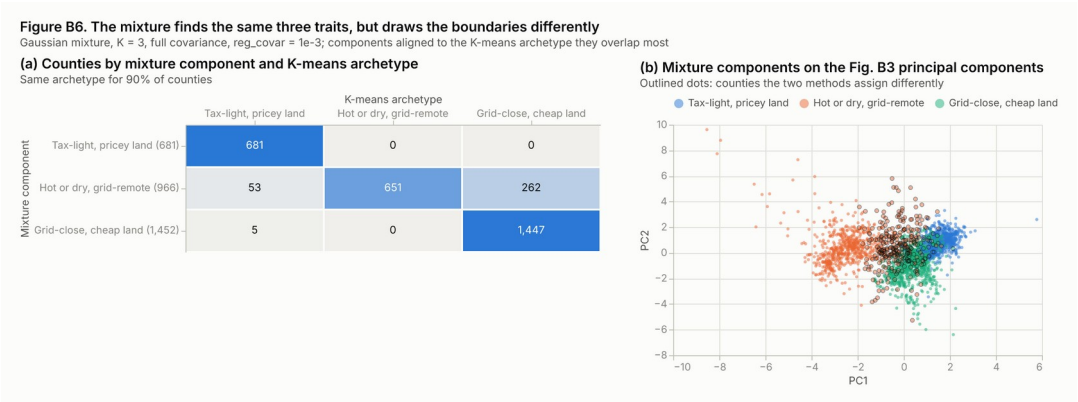


Figure B6. The mixture at K = 3 finds the same three traits. It assigns 90% of counties to the same archetype; the main difference is 262 grid-close counties that it moves into the hot-or-dry component.

Membership uncertainty. The mixture is almost never unsure: 99.4% of counties have normalized entropy below 0.1 (a top membership probability above 0.98), and even reg_covar = 0.1 leaves 96% near-certain, because near-binary features make components very sharp. We therefore measure borderline counties with the K-means distance ratio (nearest center ÷ second nearest), which does carry signal: the most borderline third of counties is twice as likely to host a data center as the third nearest a center (11.4% vs 5.6%).

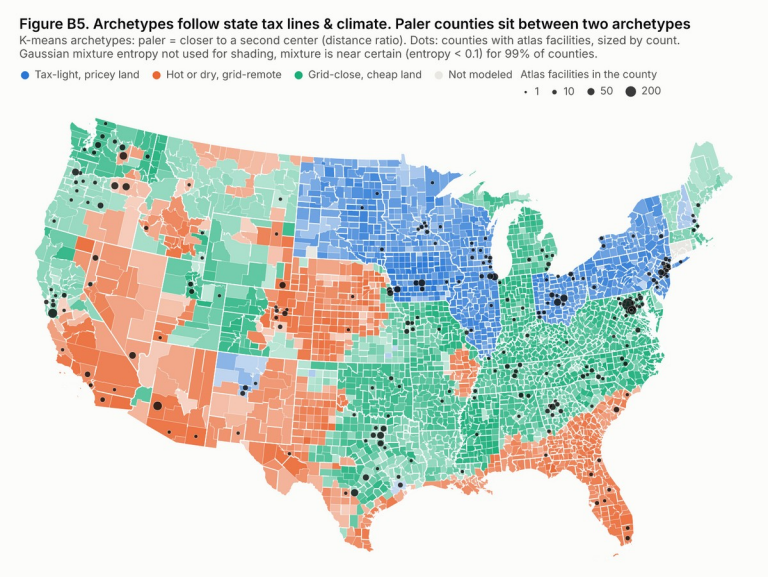


Figure B5. K-means archetypes, faded by the distance ratio, with counties that hold atlas facilities. Archetypes follow state tax lines and climate.

Table B2. Data center prevalence by archetype (7.9% of all counties host a facility; 2.8% of counties in PNNL-modeled states are chosen by its reference scenario).

Method	Archetype	Counties	With a data center	Chosen by PNNL	PNNL sites
K-means	Tax-light, pricey land	739	10.3%	2.9%	72
K-means	Hot or dry, grid-remote	651	6.1%	1.2%	40
K-means	Grid-close, cheap land	1,709	7.5%	3.3%	216
Mixture (K = 3)	Tax-light, pricey land	681	10.6%	3.1%	72
Mixture (K = 3)	Hot or dry, grid-remote	966	6.6%	2.0%	59
Mixture (K = 3)	Grid-close, cheap land	1,452	7.4%	3.1%	197

Archetypes separate siting outcomes only modestly: the share with a data center ranges from 6.1% to 10.6%, and NMI between archetype and label is 0.002. PNNL's reference scenario puts 216 of its 328 sites in the grid-close archetype, in line with CERF-DC's cost logic, while existing facilities favor the tax-light archetype.

Policy-augmented clustering. Adding the nine standardized policy features changes the archetypes substantially (ARI 0.30 against CERF-only K-means at K = 3) and loosens them (silhouette 0.261 to 0.156). The new groups track state lines more as K grows (NMI with state 0.318 at K = 3, 0.493 at K = 5) and track data centers no better (NMI with the label ≤ 0.004). Policy adds structure about states, not about siting (Appendix Table C1).

Sensitivity analysis

Table B3. Sensitivity of the clustering to outliers, initialization and the mixture's covariance floor.

Test	Change	Result
Electricity outliers	Raw vs capped at the 99th percentile	Best silhouette K = 9 → 2–3; raw data gives single-county clusters
Further outliers	Also cap land value, minimum substation distance, suitability	Sizes 664 / 1,697 / 738; ARI 0.976
Random initialization	Seven seeds (0 to 2026)	Identical inertia and sizes; ARI 1.0
Mixture covariance floor	reg_covar 0.001 → 0.1 (and 1.0)	ARI vs K-means 0.72 → 0.87, 96–99% near-certain; collapses at 1.0

Discussion

Part A: what we learned

What surprised us. First, the policy features were not ignored: they took 12% of the model's attention, more than region, yet removing them cost nothing on held-out states and shuffled profiles did nearly as well. That is what a state-level feature does when it identifies training states rather than describing siting. Second, electricity price, a core cost in CERF-DC, carries no signal on its own (PR-AUC 0.086 vs 0.079 for no skill): county-average utility rates say little about the tariffs large buyers negotiate. Third, a learned model doubles CERF's own suitability ranking (0.368 vs 0.189), and two Census demand variables would add more than all of our policy work.

What mattered most. The validation scheme, not the algorithm. Because every policy feature is identical within a state, a random split lets a model memorize “Virginia counties are positive” and credit that to policy; state-grouped CV plus a geography-only control makes a policy finding believable in either direction. The NCSL data also changed our expectations: only 16 of 1,907 bills mention “data center” and 35 carry the energy category, so “AI regulation” is a weak proxy for the policies that plausibly drive siting (utility tariffs, tax exemptions, zoning). Virginia, the largest market, has no energy, tax or data center bill in the AI database at all.

Challenges and responses. *No reusable PNNL model:* CERF-DC is a scenario-driven cost minimizer and we lacked the configurations behind its projections, so we re-framed siting as supervised learning on PNNL's own rasters, which turns its projections into an independent benchmark. *Volume and heterogeneity:* rasters of about 1.5 billion cells, a point inventory, 20 projection files and scraped HTML share no geography or schema; we profiled rasters block by block and joined everything on FIPS with validated merges. *A silent join failure:* an early notebook matched states by full name against an abbreviation column and produced zero policy values without an error; every join now reports coverage. *A late label change:* excluding facilities mapped to two counties turned Madison County, IA (a 3.7 million sq ft Microsoft campus) into a negative, so we reran every model on the final table.

With more time. (1) Build or permit dates, so policy can be tested as preceding siting. (2) County-level policy: moratoria, zoning changes and local tax abatements. (3) Demand-side features: population, fiber routes, distance to existing markets. (4) Treat the OSM-based label as positive-unlabeled, since a missing facility is not a true negative. (5) Model the 2,041-part cell-level extract instead of county averages.

Part B: what we learned

What surprised us. First, the best silhouette score was the worst clustering: K = 9 on raw data scored highest (0.399) by isolating a handful of counties with impossible electricity prices. Second, one infrastructure archetype is really a state tax regime, so policy is already inside the “infrastructure” data. Third, once outliers were capped the solution was completely stable. Fourth, the probabilistic model was the overconfident one: the mixture puts 99.4% of counties in an archetype with probability above 0.98, and it is the simpler K-means distance ratio that flags borderline counties.

Challenges and responses. Extreme raster values first appeared as a high silhouette; we traced them to 9 distinct values across 2.94 million cells in South Carolina and treated them as artifacts. Binary and near-binary features (personal property tax, cooling fractions) dominate Euclidean distance after standardization, a known weakness of K-means on mixed-type data, and make mixture components so sharp that memberships are almost all 0 or 1. A covariance floor of 0.001 kept covariances invertible but did not soften them, and BIC never reached a minimum, so we fixed K = 3 for comparison.

With more time. Cluster with a distance that treats binary features appropriately (Gower distance with K-medoids), or cluster within each tax regime so a mixture's uncertainty means something. Cluster at the cell level within high-suitability regions, and on county-level policy once it exists, since state-level policy only redraws state lines.

Tradeoffs across the project

- **Interpretability vs fit.** K = 3 is less separated (silhouette 0.26) than K = 9 on raw data (0.40), but every cluster is large enough to describe a real kind of county.
- **Coverage vs label quality.** The OSM-derived atlas is the only national open inventory, but crowd-sourced coverage means some negatives are unrecorded positives, and cleaning choices (such as dropping cross-county facilities) move individual labels.

- **Aggregation vs resolution.** County means make five sources joinable but hide the single suitable parcel that actually decides a siting.
- **Policy breadth vs relevance.** All 1,907 AI bills give every state a value; restricting to infrastructure bills is more relevant but leaves 30 of 48 jurisdictions at zero.

Ethical Considerations

Our data contain no personal information: counties, public bills and facility footprints. The main ethical risk is in deployment, because a county-level siting map is the kind of artifact used in policy, permitting and investment decisions.

Part A: deploying a siting-likelihood model

- **Regulatory arbitrage.** A model linking policy to siting can be read as a guide to which states are easiest to build in. *Mitigation:* frame results as associations, never as "policy X attracts data centers"; do not publish a ranking of states by regulatory leniency.
- **Causal misuse in lobbying.** Either result can be misused: "regulation drives data centers away" or "regulation does nothing." Our design cannot separate cause from effect, since states with data centers may legislate because of them, and we have no build dates. *Mitigation:* state the reverse-causality and timing limits next to every headline number.
- **Unequal burden on communities.** Data centers bring tax revenue but also water draw, grid-cost pass-through and noise. A likelihood map can flag counties as "next" without those communities having a voice, or help a developer target places least able to negotiate. *Mitigation:* present predictions as a preparedness tool for residents and local officials, show uncertainty, and keep the unit at the county, never the parcel.
- **Label bias.** The atlas is crowd-sourced from OpenStreetMap, which is better mapped in wealthy and urban areas, so the model may learn "well mapped" as much as "has a data center." At a 0.20 threshold it finds none of the 18 data center counties in the three smallest population quintiles, against 56% in the largest. *Mitigation:* report errors by county size and treat high-scoring negatives as possible unrecorded sites, not certain mistakes.
- **Licensing.** NCSL summaries come from LexisNexis State Net and should not be republished in raw form; the atlas is ODbL, which carries share-alike obligations for derived databases.

Part B: deploying county archetypes

- **Labels become destiny.** Naming a cluster "most suitable" can steer investment and political attention toward it, and a planner may read a description as a prescription. *Mitigation:* name clusters by measured traits ("grid-close, cheap land"), not by value judgments, and publish the profile table with every map.
- **False precision for borderline counties.** With a silhouette near 0.26, many counties sit between archetypes, and a hard color hides that. *Mitigation:* fade borderline counties, as the dashboard and Figure B5 do. We use the K-means distance ratio for this, because the mixture's membership entropy is near zero almost everywhere.
- **State boundaries presented as discovery.** One archetype is largely the set of states without a personal property tax. Presenting that as a hidden pattern would overstate what clustering found. *Mitigation:* compare clusters with plain state labels and say so when they coincide.

Statement of Work

All three members chose the datasets, explored the data and wrote the report. Contributions below follow the notebook bylines and the repository history.

Member	Contributions
Chalse Okorom-Achuonye	NCSL scraping, cleaning and bill-level feature engineering (data/ncsl_ai/pipeline/); repository owner and staged repository structure; original CERF/IM3 join pipeline and the atlas and NCSL fixes in notebook 02; policy research behind the feature definitions; co-author of notebooks 05, 08, 10 and 11; README
Khalid Jackson	CERF raster profiling and county extraction (01); cleaning and master join (02); K-means selection, outlier and seed stability, and first Gaussian mixture (03); first supervised pass with a random split (04); unsupervised lead; co-author of notebooks 05, 09 and 10
Tahreem Karim	State-grouped experimental design and shared configuration (pipeline_config.py); nested cross-validation across four model families (06); SHAP, ablation, permutation, sensitivity and failure analysis (07); PNNL benchmark (08, with Chalse); Gaussian mixture and policy clustering (09, with Khalid); report figures (10); dashboard export, design and deployment (11, 12); report writing lead

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Appendix

A. Complete feature list (3,099 modeled counties)

Feature	Group	Source raster or table	Statistic	Used in
commercial_electricity_rates_dollar_per_kwh_mean	Electricity	commercial electricity rates	county mean; capped at 99th pct	K-means; Part A
industrial_electricity_rates_dollar_per_kwh_mean	Electricity	industrial electricity rates	county mean; capped at 99th pct	K-means; Part A
composite_siting_suitability_mean	Grid and suitability	composite siting suitability	share of suitable cells	K-means; Part A
composite_siting_suitability_max	Grid and suitability	composite siting suitability	any suitable cell (0/1)	Part A
distance_to_substation_mean	Grid and suitability	distance to substation	county mean (log in Part A)	K-means; Part A
distance_to_substation_min	Grid and suitability	distance to substation	county minimum (log in Part A)	K-means; Part A
land_value_dollar_per_sqft_mean	Land and taxes	land value	county mean (log in Part A)	K-means; Part A
real_property_tax_rate_mean	Land and taxes	real property tax rate	county mean	K-means; Part A
personal_property_tax_rate_mean	Land and taxes	personal property tax rate	county mean (0 or 0.0125 by state)	K-means; Part A
sales_tax_rate_mean	Land and taxes	sales tax rate	county mean	K-means; Part A
mechanical_cooling_fraction_mean	Cooling	mechanical cooling fraction	county mean	K-means; Part A
water_cooling_fraction_mean	Cooling	water cooling fraction	county mean (= 1 − mechanical)	K-means only
pol_* (9 features)	AI policy	NCSL bills	state aggregates (Table 2)	Part A; policy clustering
div_* (9 indicators)	Geography	Census population estimates	Census division, one-hot	Part A
population_2024, pop_density	Demand	Census 2024 estimates and boundaries	county total; people per km²	Failure analysis, Table A4 only

Median versions of each mean, the state siting region and the market-location raster are kept in the master table but not modeled: medians duplicate the means, the region is an identifier, and the market raster is zero everywhere at county level and would leak the label. The master table's two raw NCSL counts are replaced by the nine policy features.

B. Noisy and missing data

Problem	Evidence	Handling
Impossible commercial electricity rates	13 South Carolina counties at 73.80–215.47 USD/kWh; 9 distinct values across 2.94 million cells	Capped at the 99th percentile (0.171 USD/kWh); 31 counties
Impossible industrial electricity rates	One Alabama county at 300.04 and one Colorado county at 5.14 USD/kWh	Capped at the 99th percentile (0.106 USD/kWh); 30 counties
Market-location raster	County majority is 0 for every county; it encodes existing facilities	Excluded (label leakage)
Missing CERF values	35 counties without values (Alaska 30, Hawaii 5); 20 without a CERF row (Connecticut 8, Alaska 5, Virginia 2, one each in CO, MT, NV, SD, and the malformed 00000 row). Connecticut's FIPS table uses legacy counties; CERF uses planning regions	Dropped; 3,099 counties modeled
Cross-county and unmatched atlas rows	5 facility IDs mapped to two counties each (10 rows); 7 rows with codes outside the reference (Connecticut planning regions, Puerto Rico)	Logged to review files; excluded from counts (1,462 rows used)
Water vs mechanical cooling	The two fractions sum to 1	Supervised models keep only mechanical cooling

C. Clustering with and without policy features

Table C1. Normalized mutual information (NMI) with state, Census division and label.

Features	Method	K	Silhouette	ARI vs CERF K-means	NMI state	NMI division	NMI label
CERF only	K-means	3	0.261	1.000	0.307	0.231	0.002
CERF only	Mixture	3	0.210	0.717	0.283	0.231	0.002
CERF + policy	K-means	3	0.156	0.303	0.318	0.163	0.002
CERF + policy	K-means	4	0.151	0.340	0.440	0.215	0.001
CERF + policy	K-means	5	0.170	0.437	0.493	0.320	0.004
CERF + policy	Mixture	3	0.122	0.238	0.291	0.157	0.002

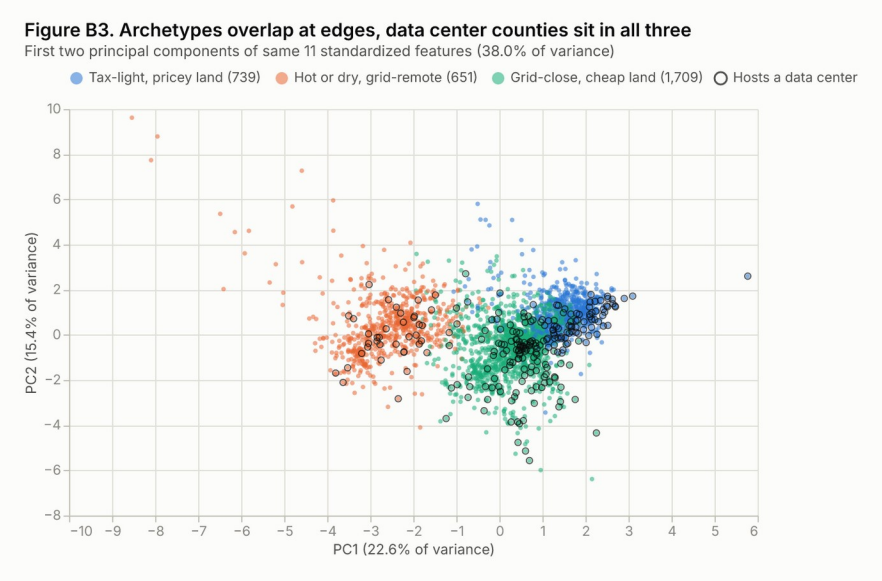


Figure C1. K-means archetypes on the first two principal components (38.0% of variance). Archetypes overlap at their edges, and data center counties sit in all three.

D. Failure hypotheses

Table A6. The six failure hypotheses fixed before the models ran, tested on all 3,099 out-of-fold predictions.

Hypothesis	What we found
Label noise (OSM atlas misses facilities)	Partly. The 20 highest-scoring counties without a facility are large (median population 419,324; 17 in the largest quintile). Review candidates, not proven misses.
Policy is constant within a state	Confirmed as a mechanism, minor as an error source: in the median state 80% of counties shift the same way when policy is added, but only 4.3% of residual variance lies between states.
Missing demand features	Confirmed, and the main failure. The largest population quintile is under-predicted (observed 30.5%, predicted 20.6%) and the three smallest over-predicted (1.0% vs 4.0%); population and density lift PR-AUC from 0.371 to 0.522.
Rural hyperscale outliers	Confirmed: 35 of the 52 counties with a facility of at least 1 million sq ft are missed at a 0.20 threshold, 18 of them in counties under 100,000 people.
Temporal mismatch	Not testable without build or permit dates.
Raster artifacts	Not material. The 15 artifact counties (2 with a facility: Aiken and Berkeley, SC) have a larger mean error (0.185 vs 0.122), but electricity alone scores only 0.086 PR-AUC and trees ignore the raw values (Table A4).

E. Repository map

Path	Purpose
data/ncsl_ai/pipeline/stage1–3	Scrape, clean and engineer NCSL bills → data/ncsl_ai/output/
scripts/01, 02	CERF raster extraction; cleaning and master join → data/cerf_siting/output/stage1–2
scripts/05	State policy features, Census division, model table → stage3_model_table/
scripts/06, 07	Nested state-grouped models; SHAP, ablation, permutation, sensitivity, failures → stage4_supervised/
scripts/08	PNNL benchmark → stage6_pnnl/
scripts/09	K-means refit, Gaussian mixture, uncertainty, policy clustering → stage5_unsupervised/
scripts/10, 11, 12	Report figures → figures/; dashboard outputs → model_outputs/; dashboard refresh → index.html
scripts/limited_policy_models/03, 04	K selection and first mixture (03); first random-split supervised pass (04)
scripts/pipeline_config.py, requirements.txt	Shared feature lists, preprocessing and folds; pinned package versions